BIOHACK 2026 · PUBLIC HEALTH & EPIDEMIOLOGY

HeatSignal NYC

A heat warning set by when New Yorkers actually get sick, and where help should go first.

[Team name] · [Member 1] · [Member 2] · [Member 3] · [Member 4]

THE PROBLEM

56%

of 7,619 heat-related ER visits in NYC from 2017 to 2026 happened on days with **no heat advisory**.

Heat kills about 500 New Yorkers each summer. Black New Yorkers die of heat stress at twice the rate of white New Yorkers.

Sources: NYC Health Dept. ER syndromic surveillance, May to Sept. 2017 to 2026 (EHDP-data); NYC Health Dept. 2026 Heat-Related Mortality Report; NWS advisory criteria

The rule misses bad days, especially early in summer

61 of 158

high-burden days (13+ heat ER visits) got no warning

+35%

more heat ER visits at 91 to 97°F before the summer's 5th hot day, when bodies have not adjusted

The 95 to 100°F range matches a 2010 study of 1997 to 2006 deaths, which advised re-checking it with new data. Early heat waves are about twice as deadly nationally.

Sources: NYC Health Dept. EHDP-data; Metzger et al., Environ Health Perspect 2010; Anderson & Bell, Environ Health Perspect 2011

Two tiers, matched to the cost of being wrong

No alert

Expected heat ER visits stay low. Routine summer operations.

Heat Watch (new)

9.35+ expected ER visits, about 20 days a summer. Calls to older adults living alone, texts, hospital heads-ups and longer cooling-center hours in 81 high-risk neighborhoods.

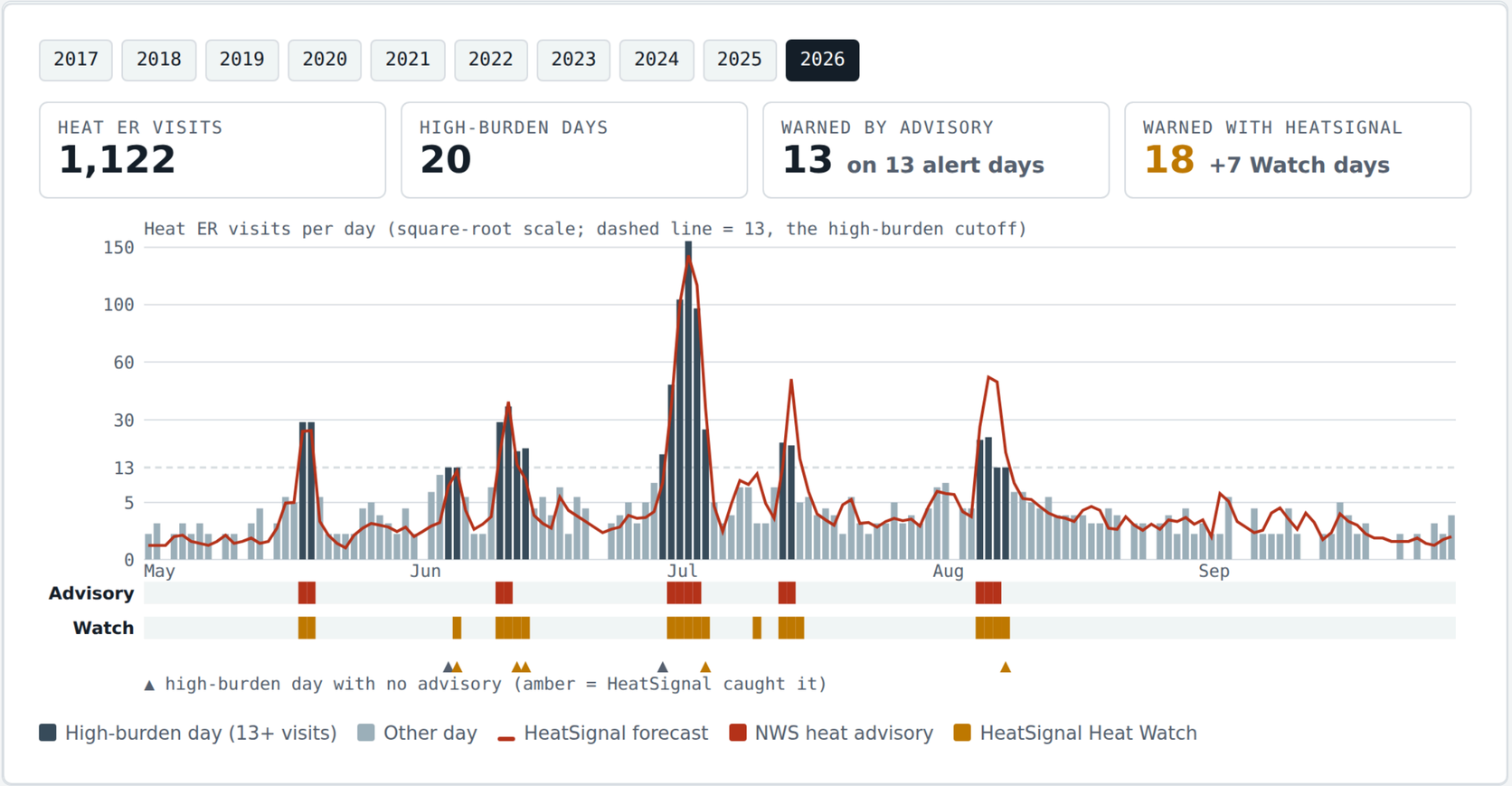
Heat Advisory

Unchanged NWS rule: heat index 100°F, or 95°F two days running. Citywide Heat Emergency Plan.

A false Watch costs some phone calls. A missed heat day can cost a life.

Already exists: NWS HeatRisk (national, climate-based, not tuned to NYC ER data) · Be a Buddy volunteers (no day trigger) · Con Edison pauses shutoffs at a 90°F forecast

Demo: replaying summer 2026, the most heat ER visits since 2017



Red line: forecast from a model that never saw 2026.

Advisory warned on 13 of 20 bad days.

With HeatSignal: 18 of 20, adding 7 Watch days.

Live prototype: biohack/index.html. Runs in any browser, loads the live NWS forecast, and lists every source.

Tested on 10 summers, each one held out

Warning rule	Alert days	Bad days warned (95% CI)	Visits covered
Current NWS advisory	118	97 · 61% (51–74%)	44%
Lower heat-index cutoff	210	129 · 82%	57%
HeatSignal: Watch + Advisory	**200**	**131 · 83% (75–91%)**	**58%**

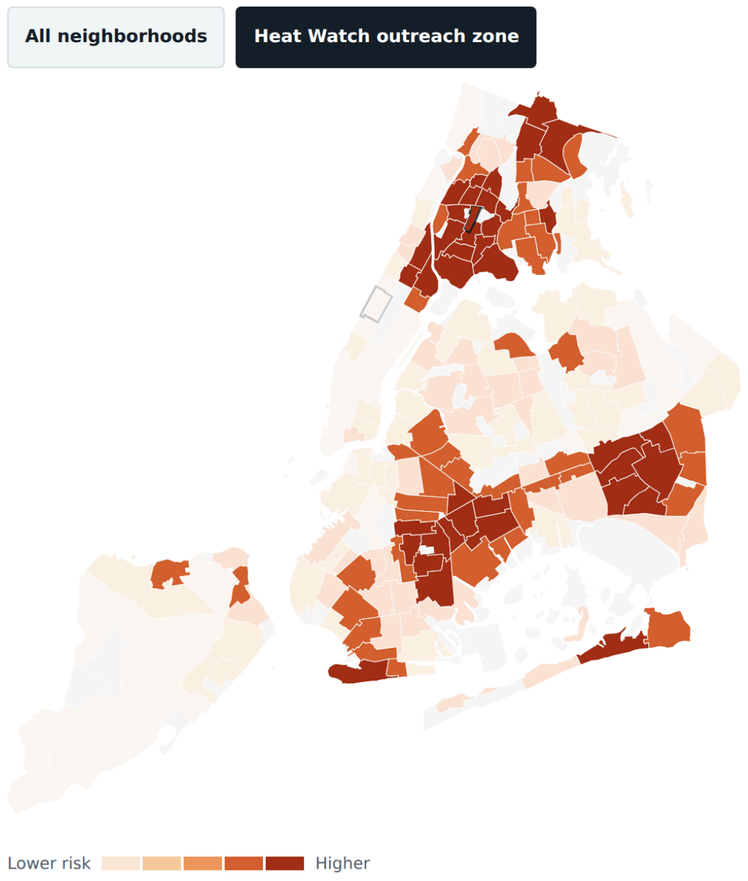
46 vs. 30 of 62 heat episodes had a warning from the first bad day.

About 1,000 more heat ER visits fell on warned days, roughly 100 a summer.

82% of bad days still caught with ±2.5°F forecast error.

Leave-one-summer-out validation; cutoffs set on training summers; 95% CIs from resampling whole summers. All 50 numbers re-checked by script (FACTS.md).

On a Watch day, help goes to 81 neighborhoods first



BX03

Claremont Village-Claremont (East)

HEAT WATCH OUTREACH ZONE

Heat vulnerability index	5 of 5
Homes with air conditioning	75.8%
Summer surface temperature	87.7°F
Green space	21.9%
Median household income	$31,144
Older adults who live alone	48%

Call list for outreach teams

Highest-index neighborhoods, ordered by fewest homes with air conditioning, then most older adults living alone.

1. Claremont Village-Claremont (East) 75.8% AC
2. Morrisania 75.8% AC
3. Tremont 75.8% AC
4. West Farms 75.8% AC
5. Crotona Park East 75.8% AC
6. Belmont 77.4% AC
7. Mount Hope 79.7% AC
8. University Heights (South)-Morris Heights 79.7% AC
9. Brownsville 81.7% AC
10. Ocean Hill 81.7% AC

Heat Vulnerability Index 4 or 5, out of 198 neighborhoods. It combines surface heat, green space, home AC, income and the share of Black residents.

The call list ranks them by fewest homes with AC, then most older adults living alone.

Top of the list: Claremont Village, Morrisania, Tremont, West Farms (Bronx), with 75.8% AC.

Source: NYC Health Dept. Heat Vulnerability Index, 2020 neighborhoods (EHDP-data)

Risks, and the road to deployment

Risks we know about

- ER counts miss deaths at home and people who avoid ERs, so harm is undercounted.
- It underpredicts the worst early waves: June 25, 2025, 31 expected, 141 actual.
- More alerts risk fatigue, so Watch goes to outreach teams, not public sirens.
- One weather station; citywide outcomes only.

Roadmap

- Now: refit each fall on the Health Department's daily counts. No new data system.
- Summer 2027: pilot with a South Bronx outreach group, tracking calls made and ER visits.
- Next: ZIP-level forecasts, overnight heat and ozone.
- Scale: 85% of US ERs report to CDC syndromic surveillance, so other cities can reuse it.

Warn when people get sick, not when a number is crossed.

Built from public data, with no new data system. The city already receives these ER counts every day.

Sources

- NYC Health Dept. Environment & Health Data Portal data: daily heat-related ED visits and max heat index at LaGuardia, 2017 to 2026; Heat Vulnerability Index. github.com/nychealth/EHDP-data
- NYC Health Dept. 2026 Heat-Related Mortality Report. a816-dohbesp.nyc.gov/IndicatorPublic/data-features/heat-report
- National Weather Service New York. Heat Advisory criteria. weather.gov/okx/extremeheat
- Metzger KB, Ito K, Matte TD. Summer heat and mortality in New York City: how hot is too hot? Environ Health Perspect 2010. doi:10.1289/ehp.0900906
- Anderson GB, Bell ML. Heat waves in the United States... 43 U.S. communities. Environ Health Perspect 2011. doi:10.1289/ehp.1002313
- Mathes RW et al. Real-time surveillance of heat-related morbidity. PLoS One 2017. doi:10.1371/journal.pone.0184364
- NOAA and CDC. NWS HeatRisk (2024). noaa.gov · CDC. About NSSP. cdc.gov/nssp
- Con Edison. Special Protections During Cold or Hot Weather. coned.com · Be a Buddy: NY1, Mar 25, 2022
- Code, data and fact check (50 numbers re-computed): FACTS.md and analysis/ in the project files